Daily Diary
Date: 01/04/2026

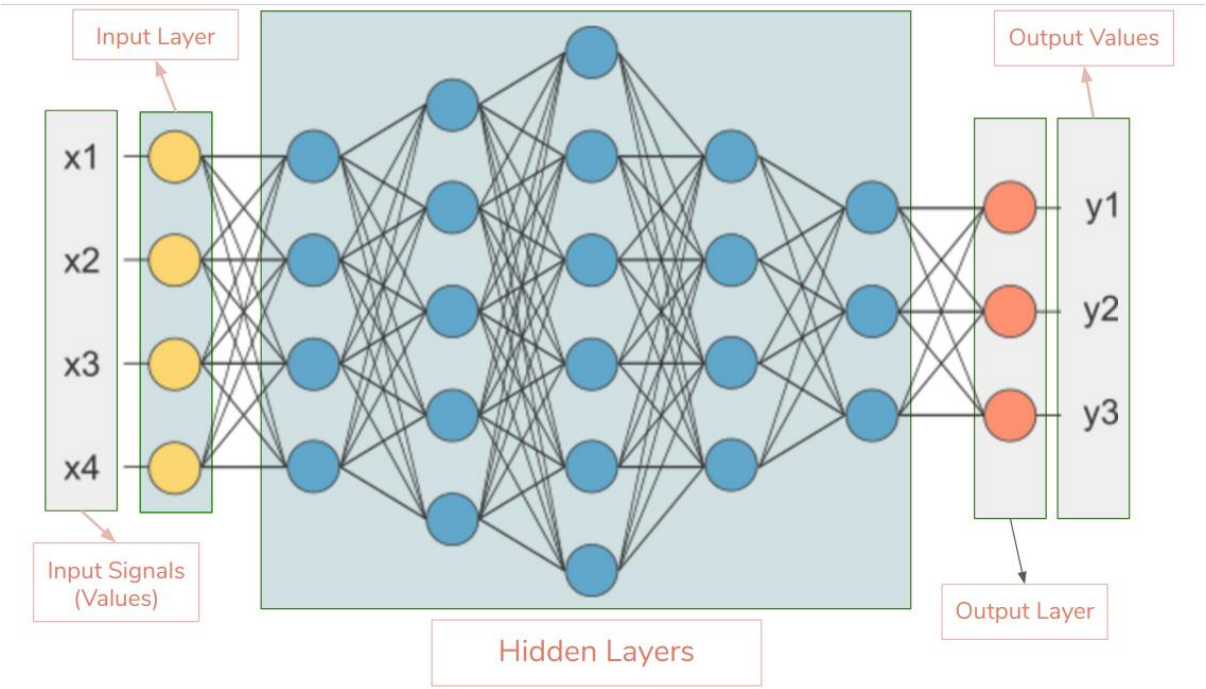
Deep Learning Fundamentals

Deep Learning (DL) is a subset of Machine Learning (ML) that utilizes artificial neural networks with multiple layers to model complex patterns in data. While it is often associated with supervised learning (using labeled data), it is also heavily utilized in unsupervised learning for tasks like feature extraction and clustering.

Neural Network Architecture

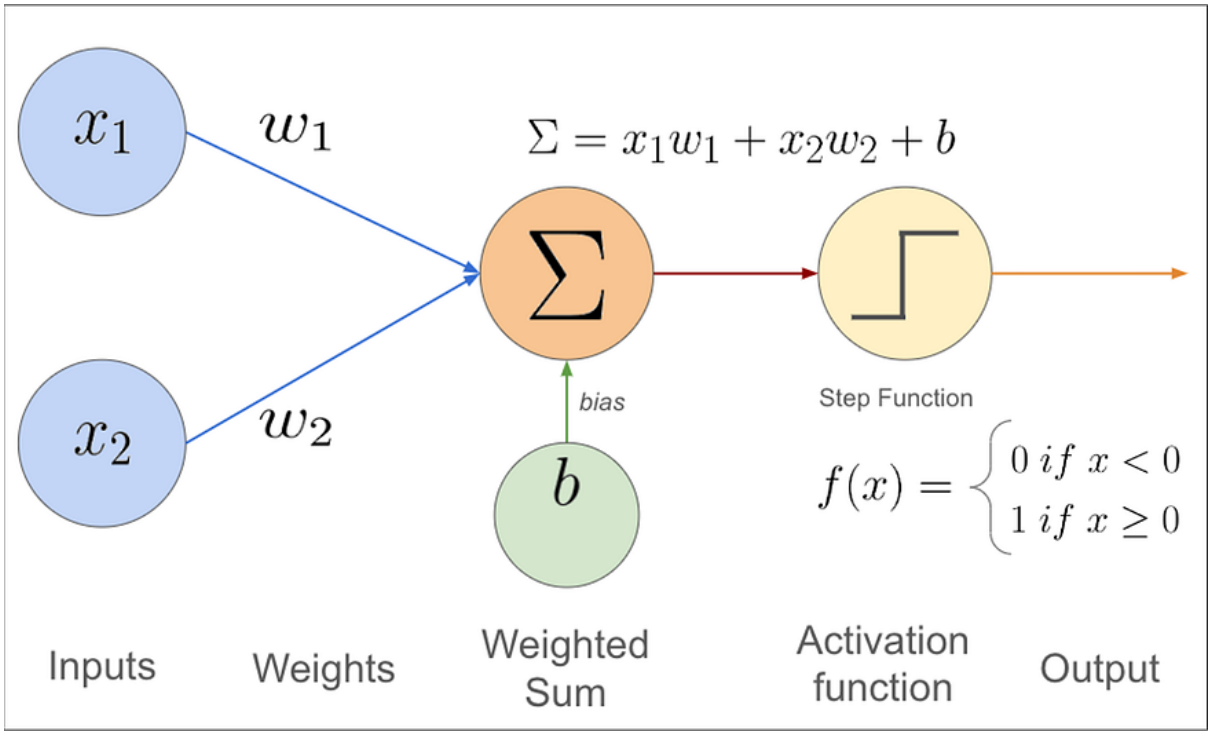
A basic neural network consists of three primary layers:

1. **Input Layer:** Receives the raw data or features.
2. **Hidden Layer(s):** Performs mathematical computations and feature extraction.
3. **Output Layer:** Produces the final prediction or classification.



Key Components

- **Perceptron:** The most basic unit of a neural network. It takes multiple binary inputs, applies weights and a bias, and produces a single output.
- **Activation Function:** A mathematical gate (e.g., ReLU, Sigmoid, Softmax) that determines whether a neuron should "fire" based on its input. It introduces non-linearity into the model.
- **Neuron Dropout:** A regularization technique where randomly selected neurons are ignored during training to prevent overfitting.

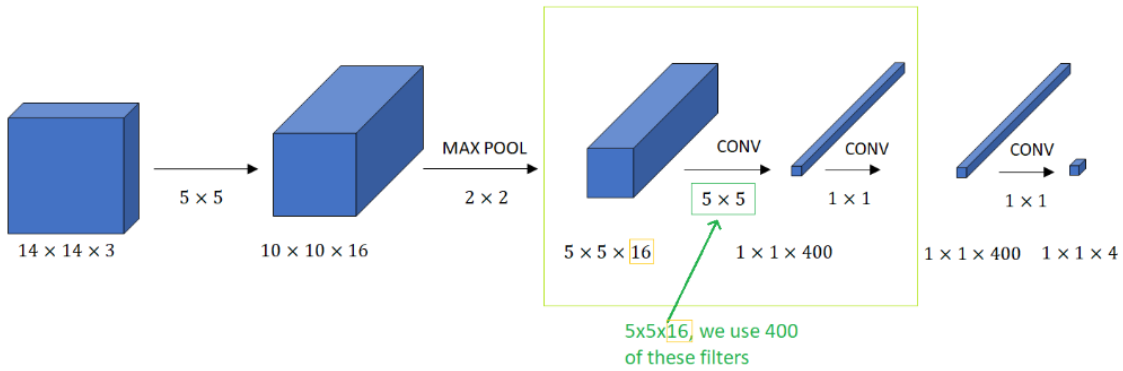


Convolutional Neural Networks (CNN)

CNNs are specialized for processing grid-like data, such as images.

CNN Operations

- **Convolution:** Images are converted into pixel grids (grayscale or RGB). Filters (kernels) slide over the image to create **Feature Maps**, which retain depth (important features) while reducing height and width.
- **Max Pooling:** A down-sampling strategy. Using a sliding window, it retains only the maximum value in a specific region (e.g., from a box of `[1, 3, 4, 8]`, it keeps `8`) to emphasize the most prominent features.



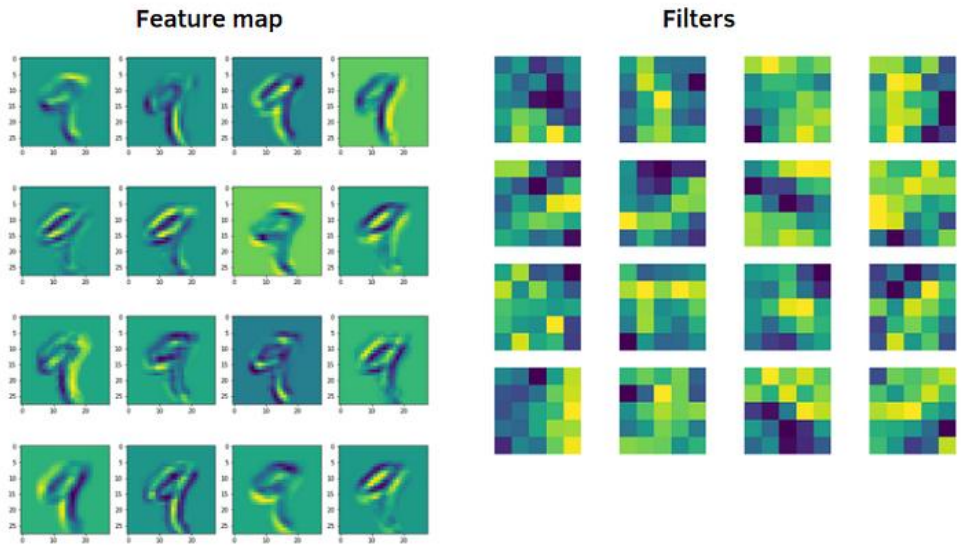
Turning Fully connected layers into Convolutional layers

- **Average Pooling:** Similar to max pooling, but calculates the average value of the window instead.

- **Flattening:** The process of converting 2D feature maps into a 1D vector so they can be fed into a standard dense neural network for final classification.

CNN Working Logic

- **Initial Layers:** Detect simple patterns like edges and lines.
- **Deeper Layers:** As the number of filters increases, the network learns to recognize complex shapes and objects.



Feature Map

In Deep Learning, specifically within **Convolutional Neural Networks (CNNs)**, a **feature map** is the output generated by applying a filter (or kernel) to an input image or the output of a previous layer.
Essentially, it is a map that highlights the locations and strengths of a specific detected feature in an image, such as an edge, a corner, or a texture.

How It Works

1. **Scanning:** A small matrix called a **filter** (e.g., 3×3 or 5×5) slides across the input pixels.
2. **Calculation:** At each position, a mathematical operation (convolution) is performed between the filter and the underlying pixels.
3. **Activation:** The result is passed through an activation function (usually **ReLU**), producing the final feature map.

Key Characteristics

- **Depth:** Each filter used in a layer creates its own unique feature map. If a layer has 64 filters, it will output 64 feature maps (often called the "depth" of the layer).
- **Dimensionality:** As the network gets deeper, feature maps typically become smaller in height and width (due to pooling) but deeper in count.

- **Abstraction:** * **Early Layers:** Feature maps detect simple things like horizontal or vertical edges.
- **Deep Layers:** Feature maps detect complex "high-level" features like eyes, wheels, or faces.

Summary Table

Step	Action	Result
Input	Raw Pixels	Grid of numbers representing color intensity.
Convolution	Filter slides over input	Mathematical extraction of specific patterns.
Feature Map	Final Output	A "filtered" version of the image highlighting specific features.